Customer Churn Prediction and Analytics: An End-to-End Data Science Approach

Project: Customer Churn Prediction & Analytics Dashboard

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Date of Submission: 20th June, 2026

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Abstract

Customer churn remains one of the costliest challenges for subscription-based businesses, particularly in the telecommunications sector. This paper documents the design, implementation, and evaluation of an end-to-end data solution that identifies customers at risk of churning, explains the drivers of attrition, and delivers actionable insights through analytical dashboards and natural-language interfaces.

Using the Telco Customer Churn dataset (7,043 customer records, 33 original features), we adopt a five-phase pipeline: (1) data cleaning and exploratory analysis in Python, (2) relational database analytics in SQL, (3) supervised machine learning for churn probability scoring, (4) interactive visualization in Power BI, and (5) a generative-AI application for business-user queries. Phase 1 analysis reveals an overall churn rate of **26.54%**, with month-to-month contract holders exhibiting a **42.7%** churn rate compared to **2.8%** for two-year contract customers. A composite payment-risk segmentation identifies a high-risk cohort with **57.7%** churn—more than eight times the rate of low-risk customers.

This document serves as the authoritative project record for all five completed phases.

1. Introduction

Retaining existing customers is significantly less expensive than acquiring new ones. In telecom, customers may leave due to competitive offers, pricing, service quality, or contract flexibility. Without a systematic approach to detect and explain churn, organizations react after revenue is lost rather than intervening proactively.

This project addresses the following research and business questions:

1. Which customers are most likely to churn?
2. What demographic, contractual, and behavioral factors drive churn?
3. How can insights be operationalized for non-technical stakeholders via dashboards and natural-language tools?

1.1 Objectives

Objective	Description
Prediction	Build models that assign a churn probability to each customer
Explanation	Identify and rank features associated with attrition
Visualization	Deliver executive and operational dashboards (Power BI)
Accessibility	Enable natural-language querying via a GenAI interface (Streamlit)

1.2 Scope and Architecture

The solution follows a layered architecture:

Data Source (Excel / SQL DB)
↓
Data Processing (Python — Pandas)
↓
SQL Layer (Customer Behavior Analysis)
↓
Machine Learning Model
↓
Visualization (Power BI)
↓
GenAI App (Streamlit + LLM)

Repository layout:

Path	Purpose
Data/Telco_customer_churn.xlsx	Raw source dataset
phase1/phase1_data_cleaning_eda.py	Phase 1 pipeline script
output/phase1/cleaned_telco_churn.csv	Cleaned, feature-enriched dataset
output/phase1/figures/	EDA visualizations
phase1/phase1_data_cleaning_eda.ipynb	Phase 1 interactive notebook
phase2/Proejct_1_Query.sql	23 business insight SQL queries
Output/phase2/Combined Query Results.csv	Combined results of SQL queries in CSV format
phase3/phase3_ml_modeling.py	Phase 3 ML pipeline script
phase3/phase3_ml_modeling.ipynb	Phase 3 interactive notebook
output/phase3/models/	Saved model artifacts (.pkl)
Phase4/Customer_Churn.pbix	Phase 4 Power BI dashboard
phase5/app.py	GenAI Streamlit application
phase5/agent.py	Gemini synthesis over verified facts
phase5/analytics_tools.py	Verified data Q&A engine (pandas)
phase5/query_insights.py	Phase 2 combined SQL results parser

Customer churn prediction has been extensively studied in marketing analytics and CRM literature. Common approaches include logistic regression as a baseline interpretable model, ensemble methods (Random Forest, Gradient Boosting/XGBoost) for improved accuracy, and model explainability techniques (SHAP, feature importance) for stakeholder trust.

Telecom churn datasets typically include demographics, service subscriptions, billing details, and a binary churn label. Key recurring findings across industry and academic studies include:

- **Contract type** is among the strongest churn predictors; month-to-month plans carry substantially higher attrition.

- **Tenure** exhibits a negative relationship with churn: longer-tenured customers are more loyal.
- **Pricing and payment method** correlate with churn; electronic check and month-to-month combinations signal elevated risk.
- **Fiber optic internet** customers often show higher churn in benchmark datasets, potentially reflecting price sensitivity or service expectations.

Our Phase 1 findings (Section 3) are consistent with this body of work and establish a foundation for modeling in Phase 3.

2. Data & Methodology

2.1 Source

- **File:** Telco_customer_churn.xlsx
- **Sheet:** Telco_Churn
- **Records:** 7,043 customers
- **Original features:** 33 columns
- **Geography:** United States (multiple cities; primarily California in sample rows)
- **Target variable:** Churn Label (Yes / No)

2.2 Variable Dictionary

Category	Variables
Identifier	CustomerID
Location	Country, State, City, Zip Code, Latitude, Longitude
Demographics	Gender, Senior Citizen, Partner, Dependents
Subscription	Tenure Months, Phone Service, Multiple Lines, Internet Service, Contract
Add-on services	Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies
Billing	Paperless Billing, Payment Method, Monthly Charges, Total Charges
Churn-related	Churn Label, Churn Value, Churn Score, Churn Reason, CLTV

2.3 Raw Data Quality Assessment

Issue	Count / Finding	Resolution (Phase 1)
Duplicate rows	0	Verified; no action required
Duplicate CustomerIDs	0	Verified; no action required
Missing Total Charges	11 (all tenure = 0)	Imputed with Monthly Charges
Missing Churn Reason	5,174 (non-churned customers)	Filled with "Not Applicable"
Redundant columns	Count (constant = 1), Lat Long (duplicate of lat/lon)	Dropped
Total Charges data type	Stored as object/string	Converted to numeric

2.4 Class Distribution

Churn Label	Count	Percentage
No	5,174	73.46%
Yes	1,869	**26.54%**

The dataset is **imbalanced** (~3:1 non-churn to churn). This has implications for model evaluation in Phase 3 (precision, recall, F1, ROC-AUC preferred over accuracy alone).

2.5 Phase 1 — Data Cleaning, Feature Engineering & EDA

Tools: Python 3.13, Pandas, NumPy, Matplotlib, Seaborn

Script: phase1/phase1_data_cleaning_eda.py

- **2.5.1 Data Cleaning Pipeline**

4. Load Excel data from Data/Telco_customer_churn.xlsx.
5. Drop redundant columns (Count, Lat Long).
6. Remove duplicate rows and duplicate CustomerIDs (none found).
7. Convert Total Charges to numeric; impute 11 zero-tenure records with Monthly Charges.
8. Fill missing Churn Reason for retained customers with "Not Applicable".
9. Strip whitespace from all text columns.

Result: Zero missing values in the cleaned dataset.

- **2.5.2 Categorical Encoding**

Variable type	Encoding scheme
Binary (Yes/No)	Yes → 1, No → 0
Gender	Male → 1, Female → 0
Ternary service fields	Yes → 1; No / No phone service / No internet service → 0
Contract	Month-to-month → 0, One year → 1, Two year → 2
Internet Service	No → 0, DSL → 1, Fiber optic → 2
Payment Method	Electronic check → 0, Mailed check → 1, Bank transfer → 2, Credit card → 3
Target	Churn Label Yes → 1, No → 0 (Churn_encoded)

Encoded columns are suffixed with _encoded to preserve original labels for reporting.

- **2.5.3 Feature Engineering**

Engineered Feature	Definition	Rationale
Tenure Group	Binned tenure: 0–12, 13–24, 25–48, 49+ months	Captures non-linear tenure effects
Avg Monthly Usage	Total Charges ÷ Tenure Months (fallback: Monthly Charges)	Proxy for usage/spend intensity

Service Count	Sum of six add-on service indicators	Measures product engagement depth
Payment Risk	High / Medium / Low composite rule	Business-actionable risk tier
Risk Segment	Low / Medium / High based on Churn Score tertiles	Aligns with existing CRM scoring

Payment Risk rule:

- **High:** Month-to-month contract AND Electronic check AND Paperless billing
- **Medium:** Month-to-month contract (other payment setups)
- **Low:** One-year or two-year contract

➢ **2.5.4 Exploratory Data Analysis**

Five visualizations were generated and saved to output/phase1/figures/:

Figure	**Description**
01_churn_distribution.png	Overall churn class counts
02_churn_by_segment.png	Churn rate by contract, internet, payment risk, tenure group
03_correlation_heatmap.png	Pearson correlation among numeric and encoded features
04_feature_importance.png	Absolute correlation with churn (importance intuition)
05_tenure_vs_charges.png	Scatter: tenure vs monthly charges, colored by churn

2.6 Phase 2 — SQL Analysis (MySQL)

Tools: MySQL 8.0

Scripts: phase2/Project_1_Query.sql

➢ **2.6.1 Database Design**

A single-table relational schema (project_1.customers) was created mirroring the Phase 1 cleaned dataset. The design uses a denormalized star-schema pattern suitable for analytical querying:

- **Primary key:** CustomerID (VARCHAR)
- **Indexes:** Contract, Internet Service, Payment Risk, Churn_encoded, CLTV, Tenure Months, City
- **53 columns** including all engineered features from Phase 1

➢ **2.6.2 Data Load Procedure**

10. Opened MySQL and selected the schema/database project_1
11. Create a table called customers with the exact columns names on the cleaned CSV file from Phase 1.
12. Executed a query to load 7,043 rows from the cleaned CSV.

➢ **2.6.3 Monthly Churn Rate — Data Limitation**

The dataset contains no calendar date column. Therefore, **Query 19 (churn rate by tenure month)** serves as the proxy for "monthly churn rate" specified in the project brief, showing at which customer-tenure month attrition peaks after signup.

➢ 2.6.4 Project 1 Query Catalog (23 Queries)

Each query in phase2/ Project_1_Query.sql includes a -- REASON: block documenting purpose, business question, and recommended action.

#	Query Title	Key Business Insight
1	Overall churn rate & customer count	Baseline KPI: 26.54% churn across 7,043 customers
2	Total revenue — retained vs churned	Quantifies historical billing revenue lost to churn
3	ARPU (overall)	Average monthly revenue per user: $64.76
4	Average CLTV by churn status	Compares lifetime value of retained vs churned
5	Churn rate by payment method	Electronic check highest friction
6	Churn rate by senior citizen status	Age-segment retention targeting
7	Churn rate by partner/dependents	Family profile loyalty analysis
8	Top 10 cities by churn rate	Geo-targeted marketing (min 30 customers)
9	ARPU by contract type	Revenue quality by commitment level
10	ARPU by internet service tier	Premium tier revenue vs churn trade-off
11	Revenue at risk (high-risk retained)	$44,485/month exposed in high-risk segment
12	High-value customer churn (CLTV > P75)	Protect top-quartile CLTV accounts
13	Total lost CLTV from churned	$7.76M total CLTV destroyed
14	Top 10 stated churn reasons	Support attitude & competitor offers dominate
15	Churn reasons by avg monthly charges	High-ARPU churn driver prioritization
16	Churn rate by service count	Engagement depth vs attrition
17	Fiber tech support impact on churn	Upsell support to reduce fiber churn
18	High-risk rule validation	Confirms month-to-month + e-check + paperless rule
19	Churn rate by tenure month	Peak-risk tenure months for touchpoints
20	Top 20 at-risk active customers (RANK)	Actionable CRM call list
21	Churn rate by charge quartile (NTILE)	Premium pricing vs attrition
22	Running lost CLTV (SUM OVER)	Pareto view of cumulative CLTV loss
23	Cross-segment: contract × internet	Fiber month-to-month worst at 54.6%

Query results exported to: output/phase2/query_results/query_01.csv … query_23.csv

A combined .csv file having all the 23 SQL queries was created for easier access.

2.7 Phase 3 — Machine Learning

Tools: scikit-learn, XGBoost, SHAP, joblib

Scripts: phase3/phase3_ml_modeling.py, phase3/phase3_ml_modeling.ipynb

➢ 2.7.1 Feature Selection

- **Target:** Churn_encoded (1 = churned)

- **22 model features:** encoded categoricals, tenure, charges, CLTV, engineered Service Count and Avg Monthly Usage
- **Excluded (leakage):** Churn Score, Churn Reason, Churn Label, Risk Segment (derived from Churn Score)
- **Excluded (raw text):** original categorical strings where _encoded versions exist

➢ 2.7.2 Modeling Pipeline

13. Stratified 80/20 train-test split (5,634 / 1,409)
14. class_weight='balanced' for Logistic Regression and Random Forest; scale_pos_weight for XGBoost
15. StandardScaler applied within Logistic Regression pipeline only
16. Models: Logistic Regression (baseline), Random Forest, XGBoost
17. Evaluation: accuracy, precision, recall, F1, ROC-AUC
18. Outputs: confusion matrices, ROC comparison, feature importance, SHAP (XGBoost), saved .pkl models

➢ 2.7.3 Model Results

Model	**Accuracy**	**Precision**	**Recall**	**F1**	**ROC-AUC**
Logistic Regression	0.744	0.511	**0.781**	0.618	**0.848**
Random Forest	0.777	0.560	0.751	0.642	0.848
XGBoost	0.774	0.558	0.706	0.623	0.838

Best model by ROC-AUC: Logistic Regression (0.848). Random Forest achieved highest accuracy (0.777) and F1 (0.642).

Churn probability scores exported for all 7,043 customers in output/phase3/churn_predictions.csv.

2.8 Phase 4 — Power BI Dashboard

Tools: Microsoft Power BI Desktop, DAX measures

➢ 2.8.1 Objective and Role in the Pipeline

Phase 4 is the **executive visualization layer** of the project. Where Phase 2 stores validated SQL metrics, Phase 3 scores individual customers, and Phase 5 answers ad hoc questions in natural language, Phase 4 provides **fixed, interactive dashboards** for leadership and operations teams who prefer visual KPI monitoring over chat or SQL.

The dashboard translates Phase 1–3 findings into at-a-glance views suitable for monthly business reviews, retention planning, and segment drill-down.

➢ 2.8.2 Data Connection

Source	**Use**
output/phase1/cleaned_telco_churn.csv	Primary data table (cleaned_telco_churn) imported into Power BI
Output/phase3/churned_predictions.csv	Secondary data table churned_predictions from phase 3 imported to Power Bi

Phase 1 engineered fields	Tenure Group, Payment Risk, Service Count, Avg Monthly Usage used in visuals and slicers
DAX measures	Computed KPIs (Churn Rate, ARPU, Revenue Lost, segment rates)

The model uses a single customer table which is a merger of the two data sources from phase1(cleaned_telco_churn.csv) and phase3(churned_predictions.csv).

- **2.8.3 Dashboard Pages and Visuals**

The report contains **four pages** and **38 visuals**:

Page	**Visuals**	**Key components**
Executive Summary	12	KPI cards (Total Customers, Retention Rate, Churn Rate, Revenue Lost, Absolute High Risk Customers); Donut chart (Churned vs Retained); Churn rate by Contract and Internet Service; Churn Rate by tenure-month line chart; Consensus Risk Donut Chart; Contract and Gender slicers
Churn Analysis	8	KPI cards (Churned Customers, MtM Churn Rate, Fiber Churn Rate, New Customer Churn Rate); Churn Rate by Payment method column chart; Top 8 Churn Reasons bar chart; Top 10 at-risk customer table; Tenure Group slicer
Revenue Insights	10	KPI cards (Total Monthly Revenue, Revenue Lost, Retained Revenue, ARPU, Avg CLTV, Revenue at Risk); Revenue Lost and Retained Revenue by Contract and ARPU by Contract column charts; Total Monthly revenue by tenure months line chart; City slicer
Customer Behaviour	8	KPI cards (Avg Tenure and Avg Charge for churned vs retained); Churn rate by City bar chart; Churn rate and Retention Rate by Tenure Group, Churn Rate by gender/senior citizen, and partner column charts

- **2.8.4 DAX Measures**

Measure	**Purpose**
Total Customers	Count of all customers (7,043)
Total Monthly Revenue	Total monthly revenue of all customers
Churned Customers / Retained Customers	Churn class counts
Churn Rate	Overall and segment churn percentage
Retention Rate	Complement of churn rate
MtM Churn Rate	Month-to-month contract churn (42.7%)
Fiber Churn Rate	Fiber optic segment churn (41.9%)
Senior Churn Rate	Percentage of senior citizen churn
New Customer Churn Rate	Early-tenure churn proxy
ARPU	Average monthly revenue per user ($64.76)
Avg Churn Probability	Average churn probability of a customer based on ML predictions

ML High Risk	Customers at high risk predicted by ML
Avg CLTV	Average customer lifetime value
Total Monthly Revenue	Sum of monthly charges
Retained Revenue / Revenue Lost	Revenue split by churn status
Revenue Loss %	Percentage of Revenue lost
Confirmed High Risk Revenue	High-risk retained customer revenue at Risk
Confirmed High Risk	Count of high payment-risk retained customers
Confirmed High Risk Rate	Percentage of customers at absolute high risk w.r.t Retained Customers.
Avg Tenure Churned / Avg Tenure Retained	Tenure comparison by churn status
Avg Charge (Churned) / Avg Charge (Retained)	Monthly charge comparison

Two new columns were added in Power BI. One is the Churn Risk Category based on the Churn Probability predicted by ML. Another is the Consensus Risk which is the comparison between the Risk Segment (column developed from Churn Score which was a target column already available in the raw dataset) and the Churn Risk Category. The Consensus Risk column was used to determine the Absolute high-risk customers i.e. the customers that have no yet churned (Churn Label = "No") but are at high risk of churning if proper action is not taken to retain them. As per the consensus risk column, there are **191** customers at absolute high risk that will cost the company a revenue of **$15,332.** These high priority customers require immediate intervention for retention.

Dashboard saved in phase4/Customer_Churn.pbix

2.9 Phase 5 — GenAI Streamlit Application

Tools: Streamlit, Google Gemini (langchain-google-genai), python-dotenv, pandas

Scripts: phase5/app.py, phase5/agent.py, phase5/analytics_tools.py, phase5/query_insights.py

2.9.1 Objective

Enable non-technical business users to ask natural-language questions about churn and receive **accurate, data-backed** answers without writing SQL or Python.

Example questions:

- "Why are customers churning?"
- "Which segment has the highest churn?"
- "Who are the top 10 at-risk customers?"
- "What is our ARPU?"
- "How does fiber optic compare to DSL for churn?"

2.9.2 Architecture (Accuracy-First)

User (browser) → Streamlit UI

↓

Verified data engine (pandas + Phase 2 SQL exports)

↓

Google Gemini (optional) — rewrites facts, never invents numbers

↓
Answer + optional chart

No API key: verified data engine returns plain-text answers directly

Design principle: All statistics are computed in Python from source data *before* any LLM is invoked. Gemini (when configured) only synthesizes natural-language responses from the verified facts block.

Data sources merged at runtime:

- output/phase1/cleaned_telco_churn.csv
- output/phase3/churn_predictions.csv (ML probabilities)
- output/phase2/query_results/Combined Query Results.csv (23 precomputed SQL query blocks)

2.9.3 Verified Data Engine

The analytical engine uses regex intent routing and pandas group-by operations. Phase 2 combined query results provide authoritative KPIs for payment method, high-risk retained customers, churn reasons, tenure trends, and cross-segment analysis.

Capability	Data source
Overall KPIs	Phase 1 CSV
Segment churn rates	Phase 1 CSV (groupby)
At-risk customer list	Phase 3 ML probabilities (fallback: Churn Score)
Payment method / high-risk / reasons	Phase 2 combined CSV
Fiber vs DSL comparison	Phase 1 CSV + Query 23 cross-segment

2.9.4 LLM Modes

Mode	Requirement	Use case
Gemini + verified data	GOOGLE_API_KEY in phase5/.env	Natural-language answers grounded in computed facts
OpenAI + verified data	OPENAI_API_KEY (optional fallback)	Same accuracy-first synthesis via GPT
Verified data engine	None	Always available; direct factual answers

2.9.5 Features Delivered

Feature	Status
Streamlit chat UI	Complete
Verified data engine (pandas + Phase 2 CSV)	Complete
Google Gemini synthesis	Complete
Phase 2 combined query results integration	Complete
Suggested question buttons	Complete
Chat history	Complete
OpenAI fallback	Complete
Optional bar charts for segment questions	Complete

Run command:

```
python -m streamlit run phase5/app.py
```

3. Results

3.1 Phase 1 — Descriptive Statistics

After cleaning and feature engineering, the dataset contains **7,043 rows × 53 columns** (20 engineered/encoded fields added).

Variable	Mean	Std	Min	Median	Max
Tenure Months	32.37	24.56	0	29	72
Monthly Charges ($)	64.76	30.09	18.25	70.35	118.75
Total Charges ($)	2,279.80	2,266.73	18.80	1,394.55	8,684.80
Churn Score	58.70	21.53	5	61	100
CLTV	4,400.30	1,183.06	2,003	4,527	6,500

3.2 Churn Rate by Contract Type

Contract	Churn Rate
Month-to-month	**42.7%**
One year	11.3%
Two year	**2.8%**

Finding: Contract commitment is the single strongest categorical differentiator. Month-to-month customers churn at more than **15×** the rate of two-year contract holders.

3.3 Churn Rate by Internet Service

Internet Service	Churn Rate
Fiber optic	**41.9%**
DSL	19.0%
No internet	7.4%

Finding: Fiber optic subscribers—despite being a premium product—exhibit the highest churn. This may reflect higher expectations, competitive fiber offers, or price sensitivity among high-speed users.

3.4 Feature Importance (Correlation with Churn)

Absolute Pearson correlation with Churn_encoded:

Feature	Correlation
Churn Score	0.665
Tenure Months	0.352
Total Charges	0.198
Monthly Charges	0.193

Avg Monthly Usage	0.193
CLTV	0.127
Service Count	0.088

Finding: The pre-computed Churn Score is highly correlated with the label (expected). Among raw behavioral features, **tenure** and **billing variables** show the strongest independent signal—consistent with prior telecom churn research.

3.5 Payment Risk Segmentation

Payment Risk	Customers	Churn Rate
High	1,397	**57.7%**
Medium	2,478	34.3%
Low	3,168	**6.8%**

Finding: The engineered payment-risk feature creates a highly separable business segment. High-risk customers (month-to-month + electronic check + paperless billing) should be prioritized for retention campaigns.

3.6 Churn Reasons (Churned Customers Only)

Among 1,869 churned customers, stated reasons include:

- Competitor made better offer
- Competitor had better devices
- Competitor offered higher download speeds
- Competitor offered more data
- Moved
- Price too high
- Poor expertise of phone support
- Lack of affordable download/upload speed
- Product dissatisfaction
- Service dissatisfaction

Finding: Competitive pressure dominates stated churn reasons, suggesting retention strategies should include competitive monitoring, loyalty pricing, and contract incentives.

3.7 Phase 2 — SQL Analysis Results

Phase 2 SQL queries were executed against the cleaned dataset. Key findings validated against Phase 1 Python EDA:

- **Executive KPIs**

Metric	Value
Total customers	7,043
Churn rate	**26.54%**
ARPU	**$64.76** / month
Total lost CLTV (churned)	**$7,755,256**
Lost monthly revenue (churned)	**$139,130.85**

- **Segmentation Highlights**

Segment	Churn Rate	Action
Month-to-month contract	42.7%	Priority upgrade campaigns
Fiber optic internet	41.9%	Fiber-specific retention program
High payment risk	57.7%	Proactive outreach to 591 retained high-risk customers
Fiber + month-to-month	**54.61%**	Worst cross-segment; targeted offers required
Two-year contract	2.8%	Model segment for loyalty programs

- **Revenue at Risk w.r.t High Payment**

Among **retained** customers, the high payment-risk segment represents **591 customers** generating **$44,485/month** in monthly charges. If this cohort churns at its historical 57.7% rate, significant revenue is at stake.

- **Top Churn Reasons**

Churn Reason	Count	% of Churned
Attitude of support person	192	10.3%
Competitor offered higher download speeds	189	10.1%
Competitor offered more data	162	8.7%
Competitor made better offer	140	7.5%
Price too high	98	5.2%

Finding: Support quality and competitive offers together account for over 30% of stated churn reasons. Investment in support training and competitive pricing intelligence is justified.

3.8 Phase 3 — Machine Learning Results

- **ROC-AUC 0.848** achieved by Logistic Regression — strong discriminative ability despite class imbalance
- **Recall 0.781** on churn class (Logistic Regression) — catches ~78% of actual churners, important for retention campaigns
- **Random Forest** offers best overall accuracy (77.7%) with balanced precision-recall trade-off
- **Churn Score excluded** from features to prevent target leakage; model relies on behavioral and contract features
- Top drivers (from feature importance): tenure, contract type, internet service, monthly charges
- All customers scored with churn probability for CRM prioritization

3.9 Phase 4 — Power BI Dashboard Results

The Customer_Churn.pbix report delivers four interactive pages validated against Phase 1–2 analytics:

Dashboard capability	Result
Executive KPI cards	7,043 customers; 26.54% churn; 73.46% retention rate
Churn segmentation	Month-to-month 42.7%; fiber 41.9%; electronic check 45.29%
Revenue metrics	ARPU $64.76; total lost CLTV $7,755,256; $44,485/month revenue at

	risk
At-risk table	Top 10 retained customers ranked by Churn Score for CRM outreach
Churn reasons	Top 8 stated reasons visualized
Behaviour analysis	Tenure and charge profiles compared for churned vs retained customers
Interactivity	Contract, Gender, Tenure Group, and City slicers with cross-filtering

3.10 Phase 5 — GenAI Application Results

The Streamlit assistant answers business questions using a verified data engine, Phase 2 SQL exports, and optional Google Gemini synthesis:

Capability	**Result**
Overall churn Q&A	Returns 26.54% rate, 7,043 customers, $64.76 ARPU
Segment analysis	Identifies month-to-month (42.7%), fiber (41.9%), high payment-risk (57.7%)
At-risk list	Surfaces top retained customers by ML churn probability
Churn reasons	Top reason: support attitude (192 customers, 10.3% of churned)
Fiber vs DSL	Fiber 41.9% churn vs DSL 19.0%; fiber ARPU $91.50
Payment method	Electronic check highest churn at 45.29% (Query 5)
High-risk retained	591 customers; $44,485 monthly revenue at risk (Query 11)
Tenure trend	Uses tenure-month proxy (no calendar dates in dataset)
Chart generation	Bar charts for segment questions

Design: The phase5 architecture computes all numbers in Python first, loads Phase 2 Combined Query Results.csv for SQL-validated KPIs, and uses Google Gemini only to rewrite verified facts into natural language.

4. Discussion

4.1 Business Implications (Phase 1)

19. **Contract conversion:** Incentivize month-to-month customers to migrate to one- or two-year plans; even a shift from month-to-month to one-year could reduce churn from ~43% to ~11%.
20. **Fiber optic retention:** Investigate fiber customer experience separately—high churn despite premium service warrants targeted quality and pricing review.
21. **High-risk payment cohort:** Proactively contact the 1,397 high payment-risk customers; offer automatic payment migration or loyalty discounts.
22. **Early-tenure focus:** With tenure negatively correlated with churn, the first 12 months are critical; onboarding and engagement programs should concentrate on new subscribers.

4.2 Limitations

- Cross-sectional snapshot; no true time-series monthly usage data (tenure month used as proxy).
- Churn Score was excluded from Phase 3 model features to prevent target leakage; it is still used in Power BI at-risk ranking as a CRM prioritization score, not a model input.
- Class imbalance requires stratified sampling and appropriate metrics (addressed in Phase 3 with balanced weights).

- Stated churn reasons are self-reported and may not capture all drivers.

4.3 Project Completion Summary

Phase	Deliverable	Status
Phase 1	Cleaned dataset, EDA, feature engineering	**Complete**
Phase 2	MySQL schema, 28 business insight queries, CSV exports	**Complete**
Phase 3	ML models (ROC-AUC 0.848), churn probability scores	**Complete**
Phase 4	Power BI dashboard (Customer_Churn.pbix)	**Complete**
Phase 5	Streamlit GenAI app with verified data + Gemini	**Complete**

5. Conclusion

Phase 1 of the Customer Churn Prediction project establishes a clean, feature-rich analytical dataset and confirms well-documented telecom churn patterns: month-to-month contracts, fiber optic service, low tenure, and high-risk payment profiles are associated with substantially elevated attrition. The overall churn rate of 26.54% represents a significant revenue retention opportunity. **Phase 2** extends these findings into a queryable MySQL analytical layer with 23 business insight queries covering executive KPIs, segmentation, revenue analysis, churn reasons, and window-function-based customer ranking. SQL analysis confirms $7.76M in lost CLTV and identifies fiber month-to-month customers (54.6% churn) as the highest-priority retention segment. **Phase 3** delivers machine learning models achieving ROC-AUC of 0.848, scoring every customer with a churn probability for proactive retention. **Phase 4** completes the executive visualization layer with a four-page Power BI dashboard (Customer_Churn.pbix) covering KPI monitoring, segment analysis, revenue insights, and customer behaviour. **Phase 5** completes the accessibility layer with a Streamlit GenAI assistant powered by a verified data engine and Google Gemini, enabling plain-English queries backed by Phase 1–3 data and Phase 2 SQL exports.

All five project phases are complete. The end-to-end pipeline—from raw Excel data through SQL analytics, machine learning, interactive dashboards, and natural-language querying—delivers a comprehensive churn prediction and retention intelligence solution for NCPL.

6. Future Scope

While the five-phase pipeline delivers a complete churn analytics solution—from cleaned data and SQL insights through machine learning, Power BI dashboards, and a GenAI query interface—several extensions would strengthen its operational value. Connecting the Power BI report and Streamlit app to a live MySQL or cloud data warehouse would enable automated refresh as new customer records arrive, replacing the current static CSV snapshot. Retraining ML models on a rolling schedule and tracking ROC-AUC and recall over time would support model governance and drift detection. Integrating churn probability scores from Phase 3 directly into the Power BI at-risk page—and pushing ranked customer lists to a CRM via API—would close the loop from insight to retention action. Where calendar dates become available, true monthly cohort and survival analysis could replace the tenure-month proxy used today. Finally, A/B testing of retention campaigns against predicted high-risk segments, combined with SHAP-based explanation in the GenAI app, would help stakeholders understand not only *who* is likely to churn but *why*, and whether interventions are working.